

Education

Rensselaer Polytechnic Institute, Troy, NY

Expected Graduation Date: May 2026

Bachelor of Science in Mechanical Engineering, Dean's Honor List, GPA: 3.99.

- 2025 Summer Undergraduate Research Award, Rensselaer Leadership Award

Research

Modification, Modeling, and Control of an Off-Road Vehicle at the Limit of Grip

Oct. 2024 to Present

Xtreme Autonomy Lab, advised by Prof. Franck Djeumou and Prof. Sandipan Mishra

- Member of the platform research team, worked on the electrical and embedded implantation of automated features, such that the platform could generate and follow aggressive trajectories based on localization and state estimation.
- Designed and manufactured many PCBs for power distribution and signal processing and spoofing. Integrated signal filters on system electronics to minimize noise and ill-effects from in-vehicle vibration and harshness.
- Designed the electronic system and housing for a 24VDC, 150A power distribution system with intelligent temperature control, safety redundancy, and an integrated status panel.
- Modelled the low-level controllers and transfer functions using a combination of physics-based vehicle dynamics and data-driven methods. Working to implement a more robust embedded implementation of low-level controllers. Augmenting low-level utility drivers of existing RTOS libraries for various serial communication protocols.

Convergent Manufacturing of Ceramic-Metal Composites

Apr. 2024 to Oct. 2024

Center for Smart Convergent Manufacturing Systems, advised by Prof. Johnson Samuel

- High-voltage and power electronics lead on the research team developing a proprietary additive manufacturing method for ceramic-metal composites (under invention disclosure stage).
- Helped to design and implement a roller-based sintering apparatus that can withstand high pressure, temperature, voltage, and current concurrently. Designed machine parts using GD&T and DFM practices.
- Helped to implement a medium vacuum system to control the atmospheric conditions for multi-field assisted laser sintering.

CRC-3 Vertical-Takeoff Fixed-Wing Aircraft

Dec. 2023 to Oct. 2024

Intelligent Systems, Automation, and Controls Lab, advised by Prof. Sandipan Mishra

- Used a combination of hand calculations, CFD, and FEA to determine the optimal rotor and wing geometries. Rapidly prototyped many versions of the wings using laser cutting, 3D printing, and CNC routing.
- Built a virtual physics model of the drone to simulate the behavior of the drone in quadcopter mode to aid the development of firmware architecture.

Publications

Ramadan, R., Yu, G. Z., Yin, P., Lee, J. J., Wong, H. Y., Wong, T. W., and Zhao, R. (2024), *Evaluating the Stability of Insulin – An X-Ray Crystallography Study*, Canadian Light Source Annual User's Meeting.

Extracurricular Experiences

MANE Student Advisory Council (Rensselaer Polytechnic Institute)

Jan. 2024 to Present

Student representative for the Department of Mechanical, Aerospace, and Nuclear Engineering

Rensselaer Motorsport (Formula SAE, Student-Built Electric Racecar Team)

Sept. 2023 to Present

- Led the design and manufacturing of chassis and suspension for RM28 (2024-2025) and RM29 (2025-2026). Led chassis and suspension manufacturing for RM27 (2023-2024).
- Combined iterative FEA with gradient-based optimization method for lightweighting of RM29's chassis. Reduced RM29's chassis weight by over 9lbs in comparison to RM28 while still achieving the torsional rigidity target.
- Practiced lightweighting while applying DFM and GD&T to create lightweight, easily manufacturable components such as rockers, wheel hubs, and centerlock wheel rims. Reduced RM29's unsprung weight by 10%. Decreased minimum machining LDR from 4.8 to 3.1 for wheel rims.
- Combined trajectory optimization with steady state cornering analysis to optimize for suspension parameters such as dynamic toe progression, heave stiffness progression due to roll, etc.
- Designed a roll-heave decoupled front suspension to enhance aerodynamic balance.

Society of Manufacturing Engineers (Rensselaer Polytechnic Institute Chapter)

Jan. 2025 to Present

Chapter board member

- 2025 Digital Manufacturing Challenge, Undergraduate Category Winner, Society of Manufacturing Engineers
- Certified Additive Manufacturing Technician

Stride Adaptive Sports – Volunteer (Snow Sports and Summer Camps)

Nov. 2024 to Present